

Binti Padaliya

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EDUCATION

New York University | Masters of Science in Computer Science

Expected: May 2027

- **GPA:** 4.0/4.0
- **Clubs and Societies:** ACM-W chapter, Programming Club, Girls Who Code

Indian Institute Of Technology Mandi (IIT) | Bachelors of Technology in Computer Science

Aug 2019 - May 2023

- **WES GPA:** 3.91/4.0 **CGPA:** 8.69/10.0
- Semester Exchange Program at RWTH Aachen, Germany (Oct 2022 - Feb 2023)

SKILLS

Languages: Python, C#, C++, Java, TypeScript, JavaScript, SQL, HTML/CSS, .NET Core, R

Frameworks & Tools: React, Node.js/Express, FastAPI, Spark, Kafka, PyTorch, TensorFlow, scikit-learn, MLflow, Triton Inference, Docker, Kubernetes, AWS, Azure, Terraform, PostgreSQL, MongoDB, Redis, MinIO, DynamoDB, REST APIs, Grafana, Prometheus

Coursework: Cloud Computing, Big Data, Advanced Deep Learning, Machine Learning Operations and Systems

WORK EXPERIENCE

Software Engineer | Microsoft, India

Jul 2023 - Aug 2025

- Built CTAF, a dashboard analyzing cross-cloud traffic to secure Microsoft's Public-to-Sovereign Azure migration; designed a Trie-based IP indexing system matching 100K+ daily IPFix records against 1M+ IPAM prefixes, cutting lookup time ~100x.
- Trained an autoencoder on engineered IPFix features to flag traffic anomalies via reconstruction error, alerting service owners for review; added background pre-computation to remove latency from user-facing calls.
- Extended CrystalNet, a network emulation platform, integrating test suites and building routing traffic simulation; built a system that maps network topology and provisions only the devices affected by a change, cutting manual setup time by 95%.

Graduate Teaching Assistant | New York University

Aug 2026 - Dec 2026

- TA for Design and Analysis of Algorithms, a graduate-level course about algorithmic complexity, DP, graphs, NP-completeness.
- Created homework assignments and led recitations, helping students bridge theoretical design with practical implementation

Software Engineering Intern | Microsoft, India

Jun 2022 - Jul 2022

- Designed an automated testing framework for the product Azure DevOps using Playwright (TypeScript), reducing manual testing effort by 70% and accelerating release cycles.
- Integrated automated test suites into CI/CD pipelines and enabling daily production-ready builds.
- Owned the project end-to-end following Agile methodology ensuring compliance, delivering a scalable solution.

Research Intern | ACS Lab, IIT Mandi, India

Dec 2020 - Jan 2021

- Developed simulated environments using **Unity** ML-Agents to benchmark Deep Reinforcement Learning algorithms(PPO, SAC)
- Analyze the effectiveness of DRL algorithms against human performance in simulated tasks.

PROJECTS

Aesthetic Hub | Python, TypeScript, NestJS, PostgreSQL, MinIO, Kubernetes, CLIP, MLP, Triton, MLflow, Argo Workflows

- Built an end-to-end personalized photo aesthetic scoring system, training per-user MLPs over 768-dim CLIP embeddings with dynamic cold-start blending, deployed on a 3-node Kubernetes cluster via Triton Inference Server.
- Designed an implicit feedback pipeline ingesting 6 interaction signal types into versioned Parquet datasets, driving weekly automated retrains via Argo Workflows with MLflow experiment tracking and quality-gated model promotion.

Trie-based IP aggregation library | C#, .NET, BenchmarkDotNet, Network Telemetry Pipelines

- Built a binary trie-based library that compresses large lists of IP address ranges into a smaller equivalent set, supporting real-time incremental updates for networking use cases like BGP routing.
- Aggregated the full IPv4 and IPv6 internet routing tables (1M+ and 235K+ entries) in 3.3s and 1.1s respectively, with individual updates completing in microseconds.

FlightFlux | Python, Spark, Kafka, AWS (EMR, MSK, S3, ElastiCache), Redis, MongoDB, Dash, AWS CDK

- Built a flight delay prediction platform combining historical flight data with live tracking to generate real-time delay risk scores.
- Trained a Random Forest model on 19M+ flight records via Spark MLlib, tuning class weights to cut false positives by ~90%; built the FastAPI serving layer and provisioned AWS infra via CDK.

AWARDS/LEADERSHIP

- Hack36 Winner 2021
- ACM-W national hackathon runner ups (COVID-19 Detection) 2020
- Merit Based Scholarship (IIT Mandi) 2022